

week-02-html

Week 02 — HTML Deep Dive

Session goal: by the end of class you have a valid, semantic profile page with a working form.

Resource: [HTML Online Editor \(Compiler, Interpreter & Runner\)](#)

HTML describes the structure of a web page. HTML elements tell the browser what each piece of content is: a heading, a paragraph, a link, a form, and so on.

What we covered (Week 01)

- Client vs. server
 - Browser and server communication
 - HTTP requests and responses
 - URL anatomy
 - HTTP methods
 - HTTP status codes
 - HTML, CSS, and JavaScript roles
 - Development environment setup
 - Node.js, npm, and Git basics
 - Extensions: Live Server + Prettier
 - Creating and serving an HTML page
 - Basic HTML document structure
 - DevTools: Elements, Console, and Network
 - Reading Network requests and status codes
 - Basic debugging habits
-

Warm-up (5 min)

Open last week's `hello-web/index.html` . Add:

1. A second paragraph about yourself
2. An unordered list of three skills
3. An image with an `alt` attribute

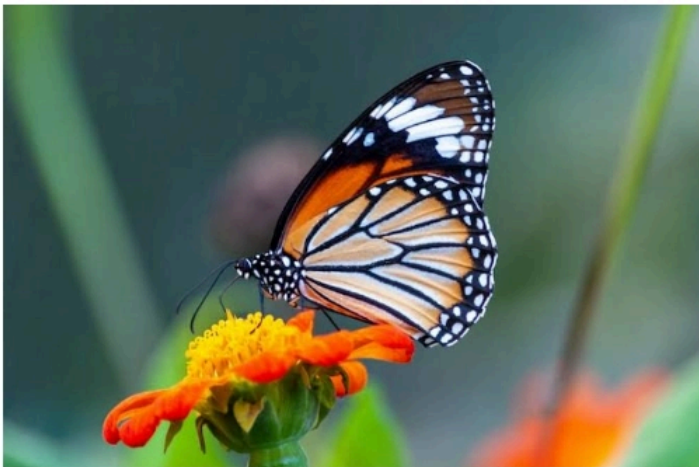
Result

Hello, Web Engineering

My name is **Your Name** and this is my first page.

I enjoy learning web development and building interactive websites.

- HTML
- CSS
- JavaScript



Click me

A Simple HTML Document

Every HTML page needs the same basic skeleton.

Example

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"/> <meta name="viewport" content="width=device-width, initial-scale=1.0"/> <meta name="description" content="A short page summary for search engines."/> <title>My Profile</title> </head> <body> <h1>Hello</h1> <p>This is my first paragraph.</p> </body> </html>
```

Example Explained

Part	Meaning
<code><!DOCTYPE html></code>	This document is HTML5
<code><html></code>	The root element of the page
<code><head></code>	Information about the page (not shown on screen)
<code><title></code>	Text in the browser tab
<code><body></code>	All visible content
<code><h1></code>	A large heading
<code><p></code>	A paragraph

Note: If you forget `charset="UTF-8"`, special characters can look broken. If you forget `viewport`, the page looks tiny on phones.

Class activity A — blank file challenge (8 min)

1. Create folder `week-02` and file `profile.html`
2. Type the skeleton from memory (doctype, html, head, body)

3. Open it with Live Server

Done when: the browser tab shows your title.

HTML Elements

An HTML element has a start tag, content, and an end tag:

```
<tagName>Content goes here...</tagName>
```

Example

```
<h1>My First Heading</h1> <p>My first paragraph.</p> <a href="https://example.com">Visit Example</a>
```

Example Explained

Start tag	Content	End tag
<h1>	My First Heading	</h1>
<p>	My first paragraph.	</p>
<a>	Visit Example	

Note: Some elements have no content and no end tag. These are called **empty elements** (void elements): `<img>`, `<br>`, `<hr>`, `<input>`, `<meta>`, `<link>`.

Example — Attributes

Attributes give extra information about an element. They go in the start tag:

```
<a href="https://example.com" target="_blank">Visit site</a>
```

- `href` = where the link goes
 - `target="_blank"` = open in a new tab
-

HTML Headings

HTML headings are defined with `<h1>` to `<h6>`.

`<h1>` is the most important. `<h6>` is the least important.

Example

```
<h1>This is heading 1</h1> <h2>This is heading 2</h2> <h3>This is heading 3</h3>
```

Example Explained

- Use **one** `<h1>` per page (the page title)
 - Do not skip levels (do not jump from `h1` to `h3`)
 - Choose headings for structure, not for font size (CSS controls size later)
-

HTML Paragraphs and Text

Example

```
<p>This is a paragraph.</p> <p>This is another paragraph.</p> <p> <strong>I  
mportant text</strong> and <em>emphasized text</em>. </p>
```

Example Explained

Tag	Use
<code><p></code>	A paragraph
<code></code>	Important text (usually bold)
<code></code>	Emphasized text (usually italic)

HTML Lists

Example — Unordered List

```
<ul> <li>Coffee</li> <li>Tea</li> <li>Milk</li> </ul>
```

Example — Ordered List

```
<ol> <li>Open the editor</li> <li>Create a file</li> <li>Save the file</li>  
</ol>
```

Example — Description List

```
<dl> <dt>HTML</dt> <dd>HyperText Markup Language</dd> <dt>CSS</dt> <dd>Cascading Style Sheets</dd> </dl>
```

Example Explained

Tag	Use
<code></code>	Bullet list (order does not matter)
<code></code>	Numbered list (order matters)
<code></code>	One list item
<code><dl></code> / <code><dt></code> / <code><dd></code>	Term and definition

HTML Links

HTML links are defined with the `<a>` tag. The `href` attribute is the destination.

Example

```
<!-- Link to another page --> <a href="about.html">About us</a> <!-- Link to another website --> <a href="https://www.w3schools.com" target="_blank" rel="noopener noreferrer"> Visit W3Schools </a> <!-- Jump to a section on the same page --> <a href="#contact">Contact</a> <!-- Email and phone --> <a href="mailto:me@example.com">Email me</a> <a href="tel:+923001234567">Call me</a>
```

Example Explained

Attribute / pattern	Meaning
<code>href="page.html"</code>	Link to another file
<code>href="https://..."</code>	Link to an external site
<code>href="#id"</code>	Jump to an element with that <code>id</code>
<code>target="_blank"</code>	Open in a new tab
<code>rel="noopener noreferrer"</code>	Safer new-tab links (always use with <code>target="_blank"</code>) <code>noopener</code> (Security Focus) and <code>noreferrer</code> (Privacy Focus): prevents a newly opened page from hacking the original tab (<code>noopener</code>) and hides the originating website's URL from analytics tracking (<code>noreferrer</code>) SEO Impact: It has zero effect on search rankings or link equity. Search engines like Google crawl and pass link authority through these tags normally

Example — Link Target

```
<section id="contact"> <h2>Contact</h2> <p>Write to me below.</p> </section>
```

The link `<a href="#contact">Contact</a>` scrolls to that section.

HTML Images

HTML images are defined with the `<img>` tag. It is an empty element (no end tag).

Example


```

```

Example Explained

Attribute	Meaning
<code>src</code>	Path or URL of the image
<code>alt</code>	Text if the image cannot load / for screen readers (required)
<code>width</code> / <code>height</code>	Size in pixels (helps stop page jump while loading)
<code>loading="lazy"</code>	Load the image only when it is near the screen

Tip: Write `alt` that describes the image. If the image is only decoration, use `alt=""`.

Example — Image with Caption

```
<figure>  <figcaption>My Week 01 hello page on Live Server</figcaption> </figure>
```

HTML Semantic Elements

Semantic elements clearly describe their meaning.

Example — Bad (no meaning)

```
<div class="header">...</div> <div class="nav">...</div> <div class="content">...</div> <div class="footer">...</div>
```

Example — Good (semantic)

```
<body> <header> <h1>My Site</h1> <nav> <ul> <li><a href="profile.html">Home</a></li> <li><a href="contact.html">Contact</a></li> </ul> </nav> </header> <main> <article> <h2>About me</h2> <p>I am learning web engineering.</p> <section id="skills"> <h3>Skills</h3> <ul> <li>HTML</li> <li>CSS</li> </ul> </section> </article> <aside> <h2>Currently learning</h2> <p>JavaScript</p> </aside> </main> <footer> <p>&copy; 2026 My Site</p> </footer> </body>
```

Example Explained

Tag	Meaning
<header>	Introductory content
<nav>	Navigation links
<main>	Main content (one per page)
<article>	Content that makes sense alone
<section>	A thematic group (usually with a heading)
<aside>	Side content (related links, tips)
<footer>	Footer for the page or section
<div>	Use only when no better tag exists

Class activity B — build the bones (12 min)

In `profile.html`, build:

1. `<header>` with your name (`<h1>`) and `<nav>` (Home / About / Contact)
2. `<main>` → `<article>` with a short bio
3. `<section>` with 4 skills in a `<ul>`
4. `<aside>` with "Currently learning..."
5. `<footer>` with copyright + `mailto:` link

Done when: View Source shows semantic tags, not only `<div>`.

Try it — Links and Images

Class activity C — media + links (10 min)

On your profile page:

1. Add one `<figure>` with `<figcaption>` and a real `alt`
2. Add three external links in the footer (all with `rel="noopener noreferrer"`)
3. Add a nav link `href="#skills"` and put `id="skills"` on your skills list

Done when: the skills link scrolls to the list.

HTML Forms

An HTML form collects user input. The `<form>` element wraps the inputs.

Example

```
<form action="#" method="post"> <label for="name">Full name</label> <input type="text" id="name" name="name" required/> <label for="email">Email</label> <input type="email" id="email" name="email" required/> <button type="submit">Send</button> </form>
```

Example Explained

Part	Meaning
<code><form></code>	Container for inputs
<code>action</code>	Where to send the data (server URL)
<code>method</code>	Usually <code>get</code> or <code>post</code>
<code><label for="id"></code>	Label linked to input with matching <code>id</code>
<code>name</code>	Name of the value when the form is submitted
<code>required</code>	Browser blocks empty submit

Important: Every input needs a `<label>` and a `name`.

Example — Input Types

```
<input type="text"/> <input type="email"/> <input type="password"/> <input type="tel"/> <input type="number"/> <input type="date"/> <input type="url"/> <input type="file"/> <input type="checkbox"/> <input type="radio"/> <input type="submit"/>
```

type	What you get
text	Normal text field
email	Checks email format + email keyboard on phones
password	Hidden characters
tel	Phone keypad on phones
number	Number field
date	Date picker
checkbox	On/off option
radio	Choose one option from a group

Example — Full Contact Form

```
<form action="#" method="post"> <div> <label for="name">Full name</label> <
input type="text" id="name" name="name" required autocomplete="name"/> </di
v> <div> <label for="email">Email</label> <input type="email" id="email" na
me="email" required placeholder="you@example.com" /> </div> <div> <label fo
r="phone">Phone</label> <input type="tel" id="phone" name="phone"/> </div>
<fieldset> <legend>Preferred track</legend> <label> <input type="radio" nam
e="track" value="frontend"/> Frontend </label> <label> <input type="radio"
name="track" value="backend"/> Backend </label> </fieldset> <div> <label fo
r="topic">Topic</label> <select id="topic" name="topic"> <option value="">C
hoose one</option> <option value="help">Help</option> <option value="feedba
ck">Feedback</option> </select> </div> <div> <label for="message">Message</
label> <textarea id="message" name="message" rows="4"></textarea> </div> <l
abel> <input type="checkbox" name="terms" required/> I accept the terms </l
abel> <button type="submit">Send</button> </form>
```

Example Explained

- `<textarea>` = multi-line text
- `<select>` + `<option>` = dropdown
- `<fieldset>` + `<legend>` = group related controls
- `placeholder` is a hint only — not a label

Class activity D — contact form (15 min)

Add a Contact section with a form with all the below inp:

- `text`, `email`, `tel`
- `select`
- `textarea`
- `checkbox`
- submit button

Then test:

1. Remove one `for` / `id` pair → label click should fail
2. Put it back
3. Submit empty → browser should block because of `required`

Done when: every label focuses its field.

HTML Tables

Use tables for **data**, never for page layout.

Example

```
<table> <caption>My tools</caption> <thead> <tr> <th scope="col">Tool</th>
<th scope="col">Comfort</th> </tr> </thead> <tbody> <tr> <th scope="row">HT
ML</th> <td>Strong</td> </tr> <tr> <th scope="row">CSS</th> <td>Beginner</t
d> </tr> <tr> <th scope="row">Git</th> <td>Comfortable</td> </tr> </tbody>
</table>
```

Example Explained

Tag	Meaning
<code><table></code>	The table
<code><caption></code>	Table title
<code><thead></code> / <code><tbody></code>	Header and body sections
<code><tr></code>	Table row
<code><th></code>	Header cell
<code><td></code>	Data cell
<code>scope="col"</code> / <code>scope="row"</code>	What the header describes

Class activity E — mini data table (5 min)

Add a small table of 3 tools/courses and your comfort level. Include `<caption>`, `<thead>`, and `scope`.

HTML `id` and `class`

The `id` and `class` attributes name elements so you can link to them, style them (Week 03), or find them with JavaScript later.

Example

```
<h2 id="skills">My Skills</h2> <p class="bio">I am learning web engineerin  
g.</p> <p class="bio highlight">HTML is my favorite so far.</p> <a href="#s  
kills">Jump to skills</a>
```

Example Explained

Attribute	Rule	Use
<code>id</code>	Must be unique on the page	One specific element (<code>#skills</code>)
<code>class</code>	Can be reused on many elements	A group of similar elements (<code>.bio</code>)
Many classes	Separate with a space	<code>class="bio highlight"</code>

Note: `id` names should not start with a number. Prefer lowercase and hyphens: `contact-form`, not `Contact Form`.

Example — Comments

HTML comments are not shown in the browser. Use them as notes for yourself.

```
<!-- This is a comment --> <!-- Main profile content starts here --> <artic  
le> <h2>About me</h2> </article>
```

Example Explained

- Start with `<!--`
- End with `-->`
- Comments help you mark sections while you build
- Do not put secret passwords in comments — anyone can View Source

Class activity F — name your sections (8 min)

On `profile.html` :

1. Give these unique `id` s: `skills` , `contact` , `about`
2. Add a nav link to each (`href="#skills"` , etc.) and click to confirm they jump
3. Add `class="card"` to your `<article>` and `class="muted"` to one paragraph (classes do nothing yet — that is fine)
4. Add three HTML comments that label: header, main content, form

Done when: clicking each nav link scrolls to the right section, and View Source shows your comments.

Lab (in class) — finish and validate

Finish `profile.html` :

- ☐ Full document skeleton (`<!DOCTYPE>` , `lang` , charset, viewport, description, title)
- ☐ Semantic tags: `header` , `nav` , `main` , `article` , `section` , `aside` , `footer`
- ☐ At least one `<figure>` + `<figcaption>` with good `alt`
- ☐ Form with 5+ input types, each with `label` + `name`
- ☐ Useful `id` s for in-page links + at least two `class` names ready for CSS
- ☐ Zero errors at validator.w3.org

Pass criteria: validator is green, and clicking a label focuses its input.

Submit: `profile.html` (+ images) and a screenshot of the validator result.

Homework

Due before Week 03.

Task 1 — Semantic teardown

Screenshot any real homepage. Draw boxes and label: `header` , `nav` , `main` , `article` , `aside` , `footer` .

Submit: annotated image.

Task 2 — Fix bad HTML

Rewrite this in `homework-rewrite.html` using semantic tags:

```
<div class="top"> <div class="big-text">My Blog</div> <div class="links"><span onclick="go('/')">Home</span></div> </div> <div class="post"> <div class="post-title">First Post</div> <div class="post-body">Hello world</div> <div class="date">Aug 9 2026</div> </div>
```

Submit: file that passes the W3C validator.

Task 3 — Second page

1. Create `contact.html` with a form
2. Link `profile.html` ↔ `contact.html` in both navs
3. Same copyright line in both footers

Submit: both HTML files.

Task 4 — Reflect (max 5 sentences)

1. Where you almost used a `<div>` and what you used instead

Common Mistakes

Mistake	Problem	Fix
Only <code><div></code> tags	No meaning for browser / screen readers	Use semantic tags